

In this page, $p < q$ are distinct primes.

Lemma. Every abelian group of order pq is cyclic.

Proof. By the Cauchy's theorem, a group G of order pq has order p element x and order q element y . Since $\gcd(p, q) = 1$, order of xy is pq , so $G = \langle pq \rangle$. $\square$

Thm. If $p \nmid q-1$, then order of pq group is cyclic.

Proof. By the lemma above, enough to show that the group is abelian.

First Proof).

By the Lagrange's Theorem, the center $|Z(G)| = 1$ or p or q or pq .

If $|Z(G)| = pq$, then $Z(G) = G$, so it is abelian.

If $|Z(G)| = p$ or q , then $G/Z(G)$ must be cyclic, so G is abelian.

Assume that $Z(G)$ is trivial. If every non-identity element of G has order p , then the class equation gives $pq = 1 + kq$,

but q does not divide 1, it is contradiction. Particular case of Cauchy's thm let $x \in G$ be the order q element and $H = \langle x \rangle$.

Then, $|G:H| = p$ is the smallest prime dividing $|G|$, so $H \trianglelefteq G$ is normal.

